

The elements of a circle and of a disc

Arcs, sectors and segments — and the angle at the centre that measures them.

LESSON 90–91

2 × 40 MIN

MATEMATIKA 6, §18

S7 8.2

LESSON CLOCK

20'

25'

25'

8'

2

Arcs and central angles	20 min
Sectors	25 min
Segments and the rest	25 min
Filling the turn	8 min
Homework	2 min

LEARNING OBJECTIVES

- Name every part of a circle and of a disc.
- Measure an arc by the angle it subtends at the centre.
- Distinguish a sector from a segment.
- Use the fact that the angles at the centre fill 360° .

TEXTBOOK

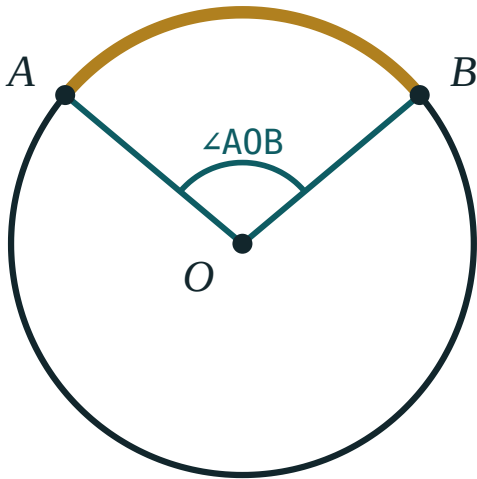
National	pp. 254–261
Cambridge	Stage 7, pp. 80–84
Workbook	Exercise 8.2

01

Explanation

Arcs and central angles

An **arc** is a piece of the curve. Its size is given by the **central angle** — the angle it makes at the centre.



a central angle equals the arc it cuts

An arc and the angle it makes at the centre

CENTRAL ANGLE	THE ARC IS	FRACTION OF THE CIRCLE
360°	the whole circle	1
180°	a semicircle	$\frac{1}{2}$
90°	a quarter	$\frac{1}{4}$
120°	a third	$\frac{1}{3}$
60°	a sixth	$\frac{1}{6}$

THE ANGLE IS THE MEASURE

Two arcs of the same circle are equal exactly when their central angles are equal. This is what makes a pie chart work: a share of the data becomes a share of 360° .

Sectors

A **sector** is a slice of the disc: the region between two radii and the arc between them.

CENTRAL ANGLE	NAME	FRACTION OF THE DISC
180°	half disc	$\frac{1}{2}$
90°	quadrant	$\frac{1}{4}$
45°	an eighth	$\frac{1}{8}$
72°	a fifth	$\frac{1}{5}$

A SECTOR IS A FRACTION OF THE WHOLE DISC

The fraction is $\frac{\alpha}{360}$. Everything about sectors — arc length, area, pie charts — comes from that one fraction.

Segments and the rest

PART	BOUNDED BY	NOT TO BE CONFUSED WITH
sector	two radii and an arc	a segment
segment	a chord and an arc	a sector
semicircle	a diameter and half the curve	a quadrant
tangent	touches the circle at one point	a chord

A SECTOR IS A SLICE OF PIZZA; A SEGMENT IS THE CRUST END

The sector reaches the centre; the segment does not. Naming them the wrong way round is the standard slip, and the picture fixes it.

Filling the turn

PROBLEM	WORKING	ANSWER
a disc split into 4 equal sectors	$360 \div 4$	90°
into 5 equal sectors	$360 \div 5$	72°
into 8 equal sectors	$360 \div 8$	45°
three sectors of 100° , 140° and x	$360 - 240$	120°

THIS IS THE PIE CHART, THREE MONTHS EARLY

Sharing 360° among categories is exactly the calculation of Quarter IV. The geometry is being built now so the statistics can use it later.

02 Worked examples

EXAMPLE 1 A disc is divided into 5 equal sectors. Find the central angle of each.

① The angles at the centre fill 360° .

② $360 \div 5$

③ $= 72^\circ$

ANSWER

72°

EXAMPLE 2 Three sectors of a disc have central angles 100° , 140° and x . Find x .

① $100 + 140 = 240$

② $360 - 240 = 120$

③ $x = 120^\circ$

ANSWER

120°

EXAMPLE 3 What fraction of a disc is a sector of 45° ?

① $\frac{45}{360}$

② Simplify by 45.

③ $\frac{1}{8}$

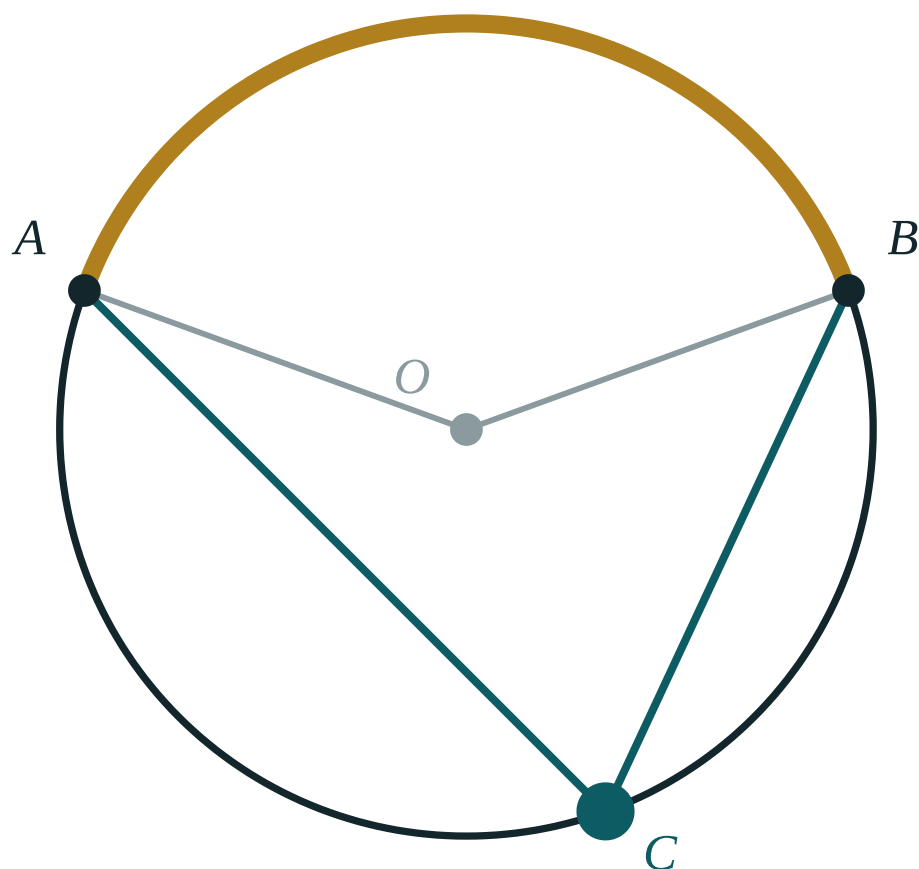
ANSWER

$$\frac{1}{8}$$

03 **Interactive model**

Cut a paper disc into sectors of different sizes and have the class name the angle of each; the fractions of 360° are then physical objects on the desk.

A sector and its central angle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Arc	Yoy	Дуга
Sector	Sektor	Сектор
Segment	Segment	Сегмент
Central angle	Markaziy burchak	Центральный угол
To subtend	Qarshi turmoq	Стягивать
Semicircle	Yarim aylana	Полуокружность
Quadrant	Chorak doira	Четверть круга
Tangent	Urinma	Касательная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An arc is measured by:

its chord

its central angle

the radius

the diameter

A sector is bounded by:

a chord and an arc

two radii and an arc

two chords

a tangent

A segment is bounded by:

a chord and an arc

two radii and an arc

two diameters

a tangent and a radius

A 90° sector is:

a semicircle

a quadrant

a segment

a third

Five equal sectors have central angles of:

60°

72°

90°

45°

A 45° sector is what fraction of the disc?

$\frac{1}{4}$

$\frac{1}{6}$

$\frac{1}{8}$

$\frac{1}{45}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A 180° arc is

ANSWER a semicircle

2. A 90° sector is

ANSWER a quadrant

3. A disc in 4 equal sectors: each angle

ANSWER 90°

4. A disc in 5 equal sectors

ANSWER 72°

5. A disc in 8 equal sectors

ANSWER 45°

6. A sector is bounded by

ANSWER two radii and an arc

7. A segment is bounded by

ANSWER a chord and an arc

Medium · the standard question

7 PROBLEMS

1. Sectors of 100° , 140° and x

ANSWER $x = 120^\circ$

2. A 120° sector as a fraction

ANSWER $\frac{1}{3}$

3. A 45° sector as a fraction

ANSWER $\frac{1}{8}$

4. A 72° sector as a fraction

ANSWER $\frac{1}{5}$

5. A disc in 6 equal sectors

ANSWER 60°

6. A disc in 12 equal sectors

ANSWER 30°

7. A tangent touches the circle at

ANSWER one point

Hard · stretch and reason

7 PROBLEMS

1. A sector that is $\frac{3}{8}$ of a disc: its angle

ANSWER 135°

2. A sector of 216° as a fraction

ANSWER $\frac{3}{5}$

3. Four sectors in the ratio $1 : 2 : 3 : 4$

ANSWER $36^\circ, 72^\circ, 108^\circ, 144^\circ$

4. Two sectors of 150° each and one more: the third

ANSWER 60°

5. A clock face: the angle between consecutive hour marks

ANSWER 30°

6. The angle swept by the minute hand in 20 minutes

ANSWER 120°

7. Why does a pie chart need 360° ?

ANSWER The whole data set fills one complete turn

07 Homework

Homework — 5 tasks

Every sector question is a fraction of 360° .

1. Draw a circle and mark on it a sector, a segment, a chord and an arc.
2. A disc is divided into 9 equal sectors. Find each central angle.
3. Sectors of 75° , 125° and x fill a disc. Find x .
4. What fraction of a disc is a sector of 108° ?

5. Find the angle of a sector that is $\frac{5}{12}$ of a disc.